

Programming Languages

Course Content

Unit	Language	Topics	Weightage
1.	C	Programming Basics (C): 1. Structure of a C program; compilation and execution 2. Data types, variables, constants, operators (arithmetic, relational, logical) 3. Input/output — scanf/printf in C 4. Type conversion; simple expressions <i>Real connect: C powers embedded systems, microcontrollers, and OS kernels.</i>	20%
2.	C	Decision Making and Control Flow (C): 1. Control Statements 2. Loops — for, while, do-while; break and continue 3. Functions — definition, call, return; call by value	20%

Structure of a C Program

A C program is generally divided into the following sections:

1. Documentation Section

Contains comments that describe the program.

```
/* Program to add two numbers */
```

2. Link Section

Contains the header files required by the program.

```
#include <stdio.h>
```

```
#include <math.h>
```

3. Definition Section

Contains symbolic constants defined using #define.

```
#define PI 3.14
```

4. Global Declaration Section

Contains global variables and function declarations that can be accessed throughout the program.

```
int x;
```

```
void display();
```

5. main() Function

The main() function is the **entry point of a C program**.

It generally contains:

- Declaration of local variables
- Executable statements
- return statement

```
int main()
```

```
{
```

```
    int a, b, sum;
```

```
    a = 10;
```

```
    b = 20;
```

```
    sum = a + b;
```

```
    printf("%d", sum);
```

```
    return 0;
```

```
}
```

6. User-Defined Function Section

Contains functions defined by the programmer.

```
void display()
```

```
{
```

```
    printf("Hello");
```

```
}
```

General Structure

```
/* Documentation Section */
```

```
#include <stdio.h>           // Link Section
```

```
#define PI 3.14              // Definition Section
```

```
int x;                       // Global Declaration
```

```
void display();             // Function Declaration
```

```
int main()                  // Main Function
```

```
{
```

```
    int a, b;               // Local Declaration
```

```
    a = 10;                 // Executable Statements
```

```
    b = 20;
```

```
    printf("%d", a + b);
```

```
    return 0;
```

```
}
```

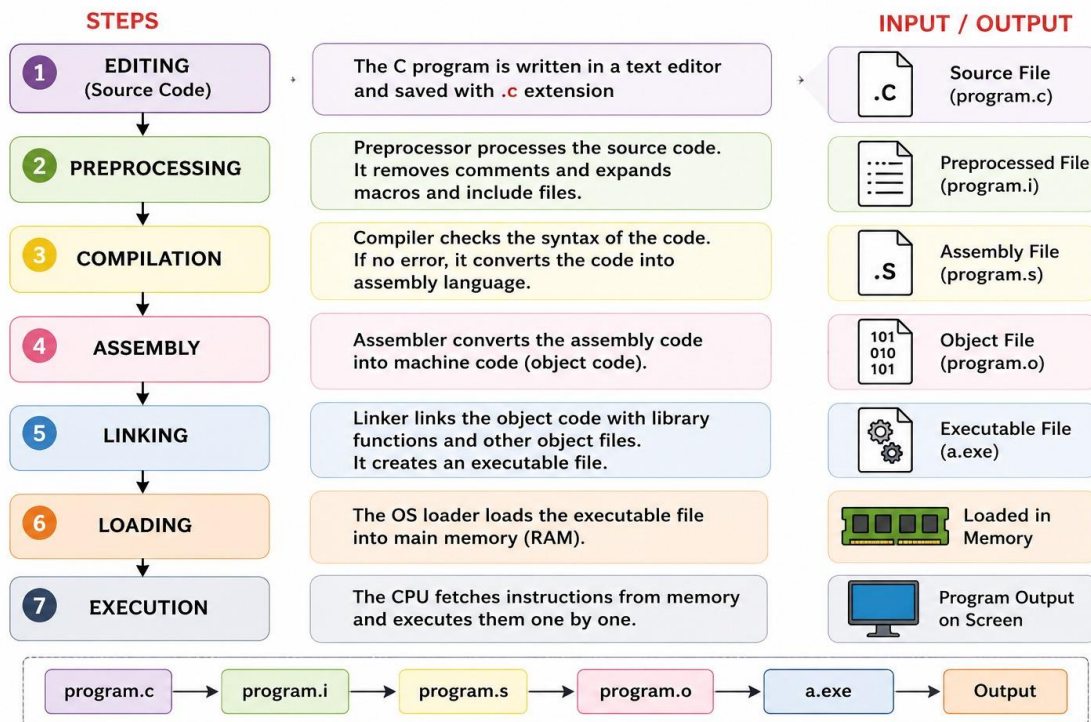
```
void display()              // User-Defined Function
```

```
{
```

```
    printf("Hello");
```

```
}
```

EXECUTION STEPS OF A C PROGRAM



Compilation and Execution in C

1. Compilation

Compilation is the process of translating a C source program into machine-understandable code through several stages.

The C source program is written in a file with .c extension.

Source Program (.c)



Preprocessor



Compiler



Assembler



Linker



Executable Program

2. Preprocessing

The **preprocessor** processes the preprocessor directives present in the source program before actual compilation.

Main functions:

- Header file inclusion using `#include`
- Macro expansion using `#define`
- Conditional compilation using `#if`, `#ifdef`, `#ifndef`, etc.
- Removal of comments

Example

```
#include <stdio.h>
```

```
#define PI 3.14
```

After preprocessing, the required header contents are included and macros are expanded.

Source Code → Preprocessed Source Code

3. Compilation

The **compiler** translates the preprocessed C program into **assembly language**.

The compiler performs:

- Lexical analysis
- Syntax analysis
- Semantic analysis
- Type checking
- Optimization
- Generation of assembly code

Preprocessed Source Code

↓
Compiler

↓
Assembly Code

The compiler also detects errors such as syntax and semantic errors.

4. Assembly

The **assembler** converts the assembly language instructions into **machine-level object code**.

Assembly Code

↓
Assembler

↓
Object Code

The object code is stored in an **object file**.

The object file contains machine instructions but may contain unresolved references to external functions and libraries.

5. Linking

The **linker** combines:

- Object files
- Required library files
- Other required program modules

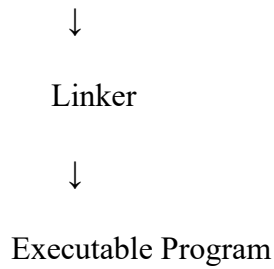
It resolves references to external functions and symbols.

For example, when a program uses:

```
printf();
```

the linker resolves the reference to the required library implementation.

Object Code + Library Code



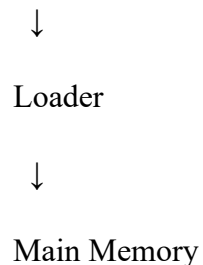
6. Loading

The **loader** is a component of the operating system that loads the executable program from secondary storage into **main memory (RAM)**.

During loading:

- The executable program is placed in memory.
- Required memory areas are prepared.
- Required dynamic libraries are loaded when applicable.
- Addresses and references are prepared for execution.

Executable Program



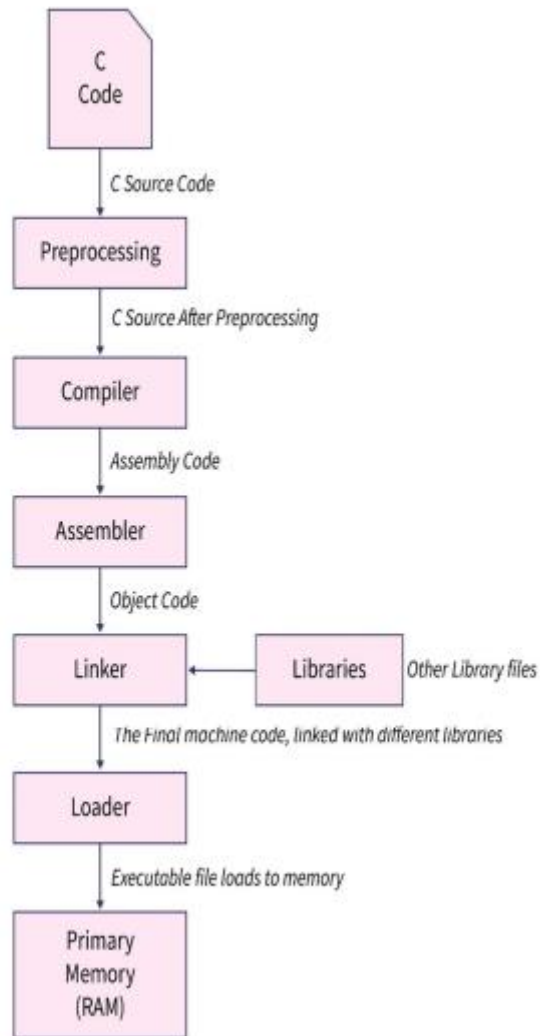
7. Execution

After the program is loaded into memory, the **CPU executes the machine instructions**.

The CPU performs the instruction cycle:

Fetch → Decode → Execute

The instructions are executed sequentially according to the program flow.



Compilation vs Execution

Compilation	Execution
Converts source code into executable/object code	Runs the executable program
Performed by compiler and related tools	Performed by CPU with operating system support
Detects compilation errors	May produce runtime errors
Occurs before execution	Occurs after loading
Produces machine-level/executable code	Produces the actual program result

Data Types, Variables, Constants and Operators in C

1. Data Types

A **data type** specifies the type of data that a variable can store.

Basic Data Types in C

Data Type	Meaning	Example
int	Integer numbers	10, -5, 100
float	Decimal numbers	10.5, 3.14
double	Large/precise decimal numbers	12.345678
char	Single character	'A', 'b', '5'

Examples

```
int age = 18;
```

```
float marks = 85.5;
```

```
double price = 1250.5678;
```

```
char grade = 'A';
```

2. Variables

A **variable** is a named memory location used to store a value that can be changed during program execution.

Syntax

```
data_type variable_name;
```

Example

```
int age;
```

```
float marks;
```

```
char grade;
```

Variable Initialization

Assigning a value to a variable at the time of declaration.

```
int age = 18;
```

```
float marks = 85.5;
```

```
char grade = 'A';
```

Changing the Value

```
int age = 18;
```

```
age = 20;
```

The value of age changes from 18 to 20.

Rules for Naming Variables

1. Variable name can contain **letters, digits and underscore** (_).
2. It cannot start with a digit.
3. Spaces are not allowed.
4. C keywords cannot be used as variable names.
5. C is **case-sensitive**.

Valid

age

student_age

marks1

total_marks

Invalid

1age

student age

float

3. Constants

A **constant** is a value that does not change during program execution.

Types of Constants

Integer Constant

10

25

-50

Floating-Point Constant

3.14

10.5

-2.5

Character Constant

'A'

'b'

'5'

String Constant

"Hello"

"Computer Science"

Using const

The const keyword is used to create a variable whose value cannot be changed.

```
const int MAX = 100;
```

```
const float PI = 3.14;
```

Example:

```
const int x = 10;
```

```
x = 20; // Error
```

4. Operators

An **operator** is a symbol used to perform an operation on one or more values.

Example:

$a + b$

Here, $+$ is an operator.

5. Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations.

Operator	Operation	Example
+	Addition	$a + b$
-	Subtraction	$a - b$
*	Multiplication	$a * b$
/	Division	a / b
%	Remainder (Modulus)	$a \% b$

Example

```
int a = 10, b = 3;
```

```
printf("%d", a + b); // 13
```

```
printf("%d", a - b); // 7
```

```
printf("%d", a * b); // 30
```

```
printf("%d", a / b); // 3
```

```
printf("%d", a % b); // 1
```

Important Point

For integer division:

$$10 / 3 = 3$$

The decimal part is discarded.

6. Relational Operators

Relational operators are used to **compare two values**.

The result is either **true (1)** or **false (0)**.

Operator	Meaning	Example
==	Equal to	a == b
!=	Not equal to	a != b
>	Greater than	a > b
<	Less than	a < b
>=	Greater than or equal to	a >= b
<=	Less than or equal to	a <= b

Example

```
int a = 10, b = 20;
```

```
printf("%d", a < b);
```

Output: 1

Because $10 < 20$ is true.

Important Difference

```
a = 10;    // Assignment
```

```
a == 10;   // Comparison
```

= is an **assignment operator**, while == is a **relational operator**.

7. Logical Operators

Logical operators are used to combine or reverse conditions.

Operator	Name	Meaning
&&	Logical AND	Both conditions must be true
!	Logical NOT	Reverses the result

Logical AND &&

(a > 5 && b > 5)

True only when **both conditions are true**.

Logical OR ||

(a > 5 || b > 5)

True when **at least one condition is true**.

Logical NOT !

!(a > 5)

Reverses the result.

Truth Table

A	B	A && B	A B
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

NOT Truth Table

A	!A
0	1
1	0

Input/Output in C — scanf() and printf()

1. scanf() Function

scanf() is a **formatted input function** used to read data from the **standard input device (keyboard)**.

It is defined in the header file:

```
#include <stdio.h>
```

Syntax

```
scanf("format_string", argument_list);
```

Example

```
int a;
```

```
scanf("%d", &a);
```

Here, %d is the format specifier for an integer and &a represents the address of variable a.

Reading Multiple Variables

```
int a, b, c;
```

```
scanf("%d %d %d", &a, &b, &c);
```

2. printf() Function

printf() is a **formatted output function** used to display data on the **standard output device (console)**.

It is defined in:

```
#include <stdio.h>
```

Syntax

```
printf("format_string", argument_list);
```

Example

```
int a = 10;
```

```
printf("%d", a);
```

3. Format Specifiers

Format specifiers specify the **type and format of data** to be read or displayed.

Format Specifier	Data Type / Use
%d	Integer
%i	Integer
%u	Unsigned integer
%f	Floating-point number
%lf	Double (scanf)
%c	Single character
%s	String
%o	Octal integer
%x	Hexadecimal integer
%X	Hexadecimal integer (uppercase)
%e	Scientific notation
%g	Fixed or scientific notation

4. scanf() with Different Data Types

Integer

```
int a;
```

```
scanf("%d", &a);
```

Float

```
float a;
```

```
scanf("%f", &a);
```

Double

```
double a;
```

```
scanf("%lf", &a);
```

Character

```
char ch;
```

```
scanf("%c", &ch);
```

String

```
char name[20];
```

```
scanf("%s", name);
```

5. printf() with Different Data Types

```
int a = 10;
```

```
float b = 12.5;
```

```
double c = 25.6789;
```

```
char ch = 'A';
```

```
char name[] = "Ankit";
```

```
printf("%d", a);
```

```
printf("%f", b);
```

```
printf("%lf", c);
```

```
printf("%c", ch);
```

```
printf("%s", name);
```

6. Format String

A **format string** contains ordinary characters and format specifiers that determine how input or output is handled.

Example:

```
printf("Age = %d", age);
```

Here:

- "Age = %d" → Format string
- %d → Format specifier
- age → Argument

7. Escape Sequences with printf()

Escape sequences are used to represent special characters.

Escape Sequence	Meaning
\n	New line
\t	Horizontal tab
\b	Backspace
\r	Carriage return
\\	Backslash
\'	Single quotation mark
\"	Double quotation mark

Example

```
printf("Hello\nWorld");
```

Output:

Hello

World

8. Width and Precision in printf()

Field Width

```
printf("%5d", 25);
```

The output occupies a field of width 5.

Precision

```
printf("%.2f", 12.3456);
```

Output:

12.35

Combined

```
printf("%8.2f", 12.3456);
```

Here:

- 8 → Field width
- .2 → Precision
- f → Floating-point value

9. Difference Between scanf() and printf()

scanf()	printf()
Formatted input function	Formatted output function
Takes input from keyboard	Displays output on console
Stores input in variables	Displays values of variables
Generally uses & for variables	Does not use & for normal variables
Example: scanf("%d", &a);	Example: printf("%d", a);

Type Conversion in C

Type conversion is the process of converting a value from **one data type to another data type**.

For example, converting an int value into a float value.

Type Hierarchy in C

For basic arithmetic type conversion, you can remember the following order from **lower** → **higher**:

Level	Data Type	Typical Size
1	char	1 byte
2	short int	2 bytes
3	int	4 bytes
4	long int	4/8 bytes
5	long long int	8 bytes
6	float	4 bytes
7	double	8 bytes
8	long double	8/12/16 bytes

Order to Remember

Char => short => int => long => long long => float => double => long double

Example

```
int a = 10;
```

```
float b = 2.5;
```

```
float c = a + b;
```

Here, int is converted to float before the addition.

Important: This is a simplified teaching hierarchy for common C expressions. “**Higher**” does **not simply mean “larger size”**; C's actual conversion rules depend on the types involved.

There are **two types of type conversion in C**:

1. Implicit Type Conversion

Implicit conversion is performed **automatically by the compiler**.

Example

```
int a = 10;
```

```
float b;
```

```
b = a;
```

Here, the int value 10 is automatically converted into float.

```
a = 10
```

```
b = 10.0
```

Example

```
int a = 10;
```

```
float b = 2.5;
```

```
float result = a + b;
```

Here, a is automatically converted from int to float.

General Rule

Smaller/Lower type → Larger/Higher type

Example:

int → float → double

2. Explicit Type Conversion (Type Casting)

Explicit conversion is performed **manually by the programmer** using a **cast operator**.

Syntax

(data_type) expression

Example

```
int a = 10;
```

```
int b = 3;
```

```
float result;
```

```
result = (float)a / b;
```

```
printf("%f", result);
```

Output:

3.333333

Here:

(float)a convert's a from int to float.

Important Example

Without Type Casting

```
int a = 5, b = 2;
```

```
float result = a / b;
```

```
printf("%f", result);
```

Output:

2.000000

Because both a and b are integers, integer division is performed.

With Type Casting

```
int a = 5, b = 2;
```

```
float result = (float)a / b;
```

```
printf("%f", result);
```

Output:

2.500000

Implicit vs Explicit Conversion

Implicit Conversion	Explicit Conversion
Done automatically	Done by programmer

Implicit Conversion	Explicit Conversion
No cast operator required	Cast operator is required
Performed by compiler	Specified by programmer
Example: float b = a;	Example: float b = (float)a;

Implicit: Compiler converts the type automatically.

Explicit: Programmer tells the compiler which type to convert using:

(data_type)

Simple Expressions in C

An expression in C is a combination of constants, variables and operators that produces a value.

Example

a + b

x * 5 + 10

(a + b) / 2

1. Operator Precedence

Operator Precedence determines which operator is performed first when an expression contains multiple operators.

Example

```
int result = 10 + 5 * 2;
```

Here, * has higher precedence than +.

So:

10 + 5 * 2

= 10 + 10

= 20

Common Precedence Order

Priority	Operators	Description
1	()	Parentheses
2	* / %	Multiplication, Division, Modulus
3	+ -	Addition, Subtraction
4	< <= > >=	Relational
5	== !=	Equality
6	&&	Logical AND
7	?:	Conditional
8	=	Assignment

Higher priority number in the table means lower precedence.

Example

```
int x = 10 + 20 * 3;
```

First:

$$20 * 3 = 60$$

Then:

$$10 + 60 = 70$$

Therefore:

$$x = 70$$

2. Associativity

Associativity determines the direction in which operators of the same precedence are evaluated.

There are two directions:

- Left to Right
- Right to Left

Example: Left-to-Right

`int x = 20 / 5 * 2;`

/ and * have the same precedence.

Their associativity is left to right.

So:

$20 / 5 * 2$

$= 4 * 2$

$= 8$

Not:

$20 / 10 = 2$

Example: Right-to-Left

Assignment operators have right-to-left associativity.

`a = b = c = 10;`

Evaluation:

`c = 10`

`b = c`

`a = b`

Therefore:

`a = 10`

`b = 10`

`c = 10`

Precedence vs Associativity

Precedence	Associativity
Determines which operator is performed first	Determines direction of evaluation
Used when operators have different precedence	Important when operators have the same precedence
Example: * before +	Example: * and / are left-to-right

Precedence → Which operator first?

Associativity → Which direction?

Example

`int x = 10 + 20 * 3 - 5;`

Step 1: * first

$10 + 60 - 5$

Step 2: + and - have same precedence and are left-to-right

$70 - 5$

$= 65$

Answer: 65

When multiplication (*) and division (/) have the same precedence, C follows left-to-right associativity.

So, whichever appears first from the left is executed first.

Example 1

$20 / 5 * 2$

Both / and * have the same precedence.

From left to right:

$$20 / 5 * 2$$

$$= 4 * 2$$

$$= 8$$

Answer = 8

Example 2

$$20 * 5 / 2$$

From left to right:

$$20 * 5 / 2$$

$$= 100 / 2$$

$$= 50$$

Answer = 50

Operator Precedence and Associativity

Operator Precedence

Operator precedence determines **which operator is evaluated first** when an expression contains different operators.

For example:

$$5 + 2 * 3$$

* has higher precedence than +, so:

$$5 + (2 * 3)$$

$$= 5 + 6$$

$$= 11$$

Operator Associativity

Associativity determines the **order of evaluation** when operators have the same precedence.

For example:

$$20 / 5 * 2$$

/ and * have the same precedence and **left-to-right associativity**:

$$(20 / 5) * 2$$

$$= 4 * 2$$

$$= 8$$

Evaluate the Expression (Imp)

Given:

$$x = 5 * 2 - 8 / 4 + 3 > 6 \&\& 1$$

We apply the operators according to their precedence.

Step 1: Multiplication and Division

* and / have higher precedence than + and -.

$$5 * 2 = 10$$

$$8 / 4 = 2$$

Therefore:

$$x = 10 - 2 + 3 > 6 \&\& 1$$

Step 2: Addition and Subtraction

+ and - have the same precedence and are evaluated **left to right**.

$$10 - 2 + 3$$

$$= 8 + 3$$

$$= 11$$

Therefore:

$$x = 11 > 6 \&\& 1$$

Step 3: Relational Operator >

$$11 > 6$$

This is **true**, so in C:

$$11 > 6 = 1$$

Therefore:

$$x = 1 \ \&\& \ 1$$

Step 4: Logical AND &&

$$1 \ \&\& \ 1 = 1$$

Therefore:

Final Answer

$$x = 1$$

Complete Order of Evaluation

$$5 * 2 - 8 / 4 + 3 > 6 \ \&\& \ 1$$

↓

$$10 - 2 + 3 > 6 \ \&\& \ 1$$

↓

$$8 + 3 > 6 \ \&\& \ 1$$

↓

$$11 > 6 \ \&\& \ 1$$

↓

$$1 \ \&\& \ 1$$

↓

$$1$$

Final value: x = 1

If-Else Statement

The **if-else statement** is a decision-making statement in C. It is used to execute one block of statements when a condition is true and another block when the condition is false.

Syntax

```
if (condition)
{
    // statements if condition is true
}
else
{
    // statements if condition is false
}
```

Example

```
#include <stdio.h>

int main()
{
    int age = 20;

    if (age >= 18)
    {
        printf("Eligible to vote");
    }

    else
    {
        printf("Not eligible to vote");
    }
}
```

```
    return 0;

}
```

Else-If Ladder

The **else-if ladder** is used when there are multiple conditions to check. Conditions are checked from top to bottom, and the block of the first true condition is executed.

Syntax

```
if (condition1)

{

    // statements

}

else if (condition2)

{

    // statements

}

else if (condition3)

{

    // statements

}

else

{

    // statements

}
```

Example

```
#include <stdio.h>
```

```
int main()
{
    int marks = 75;
    if (marks >= 90)
    {
        printf("Grade A");
    }
    else if (marks >= 75)
    {
        printf("Grade B");
    }
    else if (marks >= 50)
    {
        printf("Grade C");
    }
    else
    {
        printf("Fail");
    }
    return 0;
}
```

Important Points

- if checks a condition.
- else executes when the if condition is false.

- else if is used to check multiple conditions.
- Only the first true condition in an else-if ladder is executed.
- In C, **0 represents false** and **any non-zero value represents true**.

IF ELSE

```
#include <stdio.h>
```

```
int main() {
```

```
    int number = 70;
```

```
    if (number >= 50) {
```

```
        printf("number is an greater than 50.");
```

```
    }
```

```
    else {
```

```
        printf("number is an less than 50.");
```

```
    }
```

```
    return 0;
```

```
}
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int number = 70;
```

```
    if (number == 50) {
```

```
        printf("number is an equal to 50.");
```

```
    }
```

```
    else if (number >= 50)
```

```
    {
```

```
        printf("number is an greater than 50.");
```

```

    }

    else {

        printf("number is an less than 50.");

    }

    return 0;

}

#include <stdio.h>

int main() {

    int number = 70;


    if (number >= 50) {

        printf("%d is an greater than 50.",number);

    }

    else {

        printf("%d is an less than 50.",number);

    }

    return 0;

}

```

// Check whether an integer is odd or even

```

#include <stdio.h>

int main() {

    int number;

    if (number%2 == 0) {

        printf("%d is an even integer.",number);
    }
}

```

```
}  
  
else {  
  
    printf("%d is an odd integer.",number);  
  
}  
  
return 0;  
  
}  
  
printf("Enter an integer: ");  
  
scanf("%d", &number);
```